LiDAR–RADAR Sensor Fusion for 3D Obstacle Detection and Multi-Object Tracking (Phase 2)

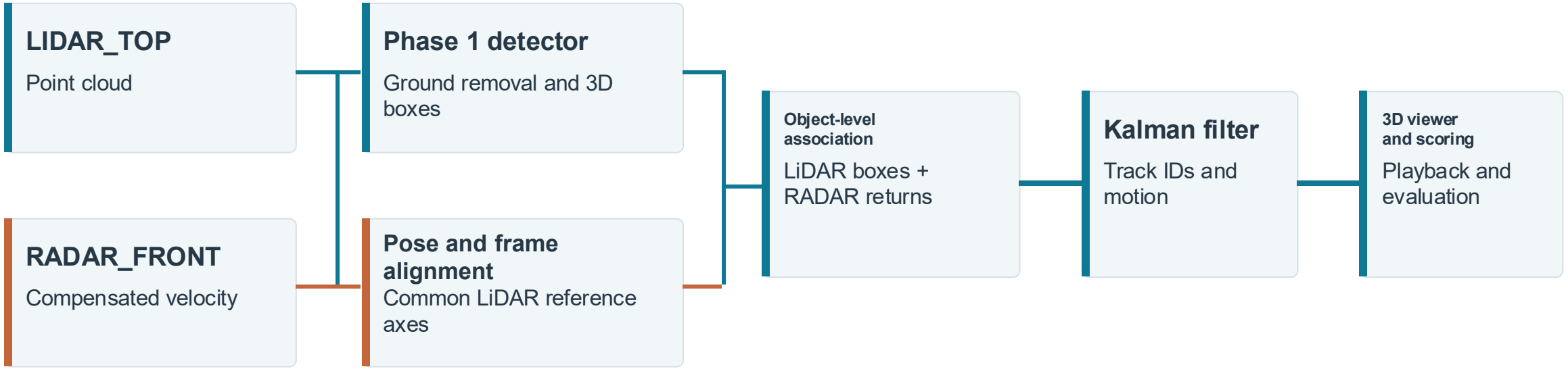
System architecture • Results • Phase 3 direction

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Project overview

Phase 2 extends the Phase 1 geometric detector with RADAR velocity and persistent multi-object tracking.

SYSTEM ARCHITECTURE



404 annotated samples

10 driving scenes

Ground truth is used for reference and evaluation, not to generate detections.

Phase 2 compares LiDAR-only and LiDAR–RADAR tracking, then uses the failure analysis to define Phase 3 work.

Methodology

LiDAR provides geometric position; RADAR contributes velocity after sensor and ego-pose alignment.

LIDAR PROCESSING

The Phase 1 pipeline remains the detection foundation.

- RANSAC ground removal; 0.15 m distance threshold
- DBSCAN: 0.40 m radius, minimum 8 samples
- Minimum 10 points per box
- Ground-model guard, size filters and selected elevated-fragment filter

RADAR PROCESSING AND ASSOCIATION

Positions and compensated velocities share the LiDAR frame.

- RADAR_FRONT calibration and ego poses align the sensor data
- LiDAR box expands 1.0 m horizontally
- A return joins the nearest eligible box, at most one
- Associated velocity vectors are averaged

KALMAN FILTER AND TRACKING

State: $[x, y, vx, vy]^T$

Constant-velocity prediction • LiDAR corrects position • RADAR corrects velocity when available

Greedy one-to-one association within 3 m; a track is removed after more than two missed updates.

Object Detection Performance

Vehicle and pedestrian annotations within 25 m; class-agnostic BEV IoU threshold 0.1.

24.0%
Precision

30.0%
Recall

26.7%
F1-score

0.378
Mean matched BEV IoU

Detection metric	Result
True positives	1,812
False positives	5,751
False negatives	4,222

The report identifies sparse LiDAR returns, clustering errors and nearby-object separation as limitations.

INTERPRETATION

The detector produces geometric candidate boxes, not predicted object classes. Precision and recall show that many annotated objects are missed and many predictions do not match.

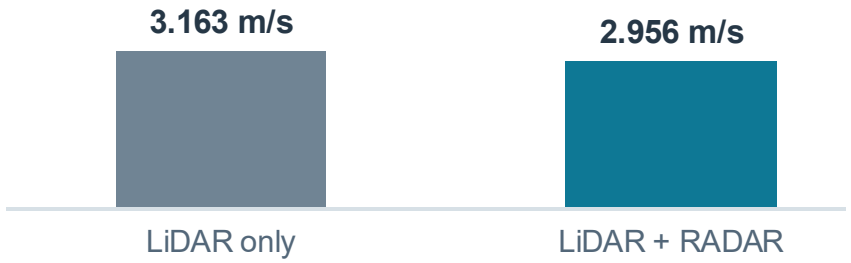
EVALUATION CAVEAT

This custom diagnostic is not the official nuScenes detection benchmark. Counts are observations across frames, not unique objects.

LiDAR-only vs LiDAR–RADAR tracking

Same recorded detections/timestamps and tracking parameters; only RADAR velocity initialization and updates are toggled.

Velocity vector RMSE



−6.5% RMSE

Conditional ID continuity



+0.85 percentage points

Tracking metric	LiDAR only	LiDAR + RADAR
Matched target observations	2,633	2,638
False track observations	9,265	9,214
Tracking precision	22.13%	22.26%
Tracking recall	43.64%	43.72%
Tracking F1-score	29.37%	29.50%
Consecutive ID changes	354	337
Mean matched position error	0.771 m	0.777 m
Matched velocity RMSE	3.163 m/s	2.956 m/s

Errors are conditional on matched tracks; the two methods can match different objects because estimated states affect association and scoring.

3.3 RADAR Association Coverage

RADAR returns provide velocity only when a return is associated with a LiDAR detection.

718

of 7,563 detection observations

9.5% associated with RADAR

This coverage was measured within the evaluated 25 m range.

Why coverage was limited

- Only the front RADAR channel is used.
- Geometric association depends on a return falling inside an expanded LiDAR box.
- Nearby objects or returns outside the box can prevent a match.

Limited association coverage helps explain the modest aggregate tracking change.

Interactive 3D Visualization

The viewer replays consecutive scans and displays sensor returns, detections, track IDs, estimated velocities and trajectories.



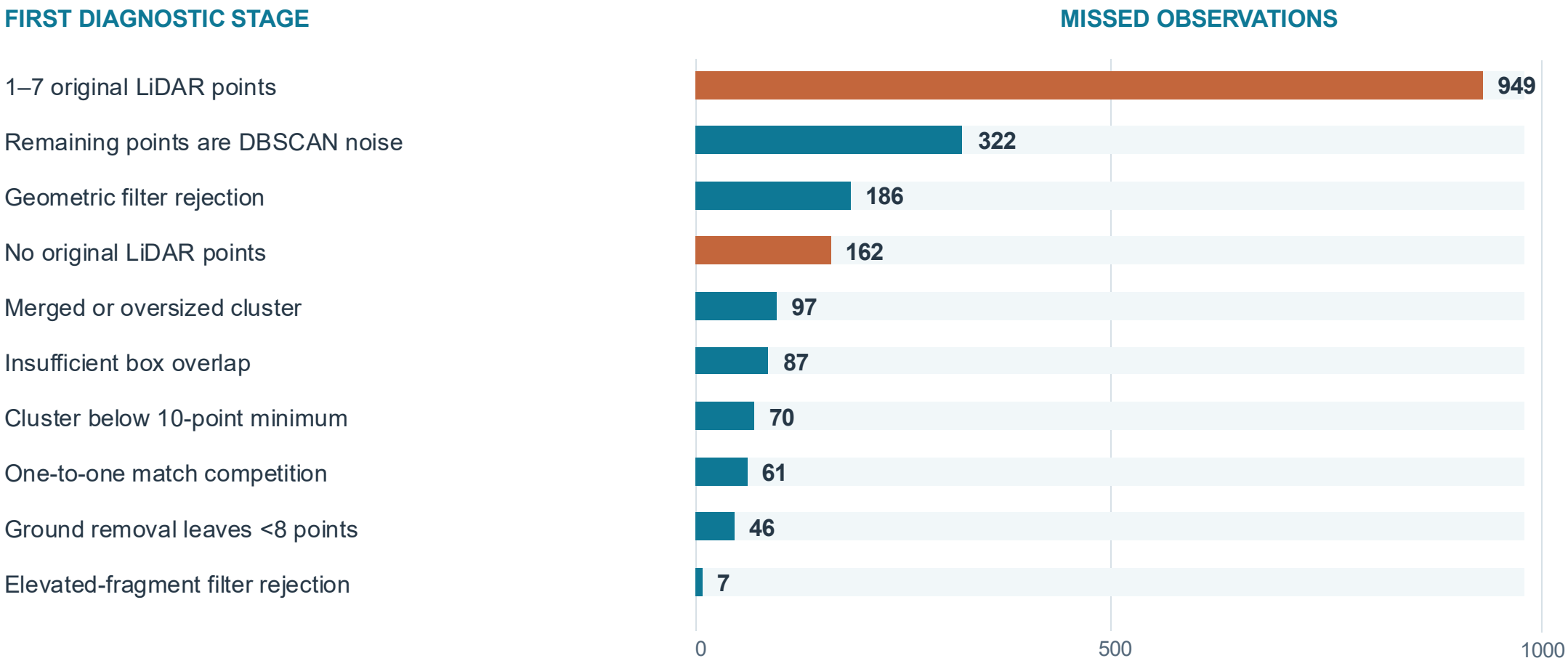
VISUAL ELEMENTS

- **Green-to-blue points**
  LiDAR distance coloring
- **Yellow / green boxes**
  Detections without / with RADAR
- **Pink points**
  RADAR returns
- **Track labels and arrows**
  IDs, Kalman speed and motion
- **Optional overlays**
  Predicted tracks and ground truth

Playback is a visualization tool, not an additional accuracy test.

Detection Limitations and Pedestrian Analysis

543 of 2,530 pedestrian observations matched; 1,987 were missed.



1,111 misses (55.9%) had fewer than 8 original LiDAR points.

Categories identify the first observed failure stage; they do not prove a unique physical cause.

Future Work (Phase 3)

The next phase should improve reliability for pedestrians and objects with sparse or ambiguous measurements.

1

Improved object detection and classification

Evaluate a pretrained 3D detector to distinguish vehicles, pedestrians and other road users.

2

Enhanced LiDAR–RADAR fusion

Add remaining RADAR sensors and investigate stronger measurement-association methods.

3

Improved clustering and object separation

Test range-dependent clustering or segmentation to separate nearby objects while preserving sparse detections.

4

More reliable tracking and evaluation

Investigate stronger association and motion models; use independent data and established benchmark protocols.

Evaluate proposed improvements individually to identify measurable benefits.

6. Conclusion

Phase 2 extends geometric LiDAR detection into a multi-sensor perception and tracking prototype.

LiDAR boxes provide object position; associated RADAR velocities support Kalman motion estimates across consecutive frames. The interactive viewer makes results and errors easier to inspect.

6.5%
lower matched velocity RMSE

+0.85 pp
conditional ID continuity

Persistent limits

False detections, missed pedestrians, merged clusters and limited RADAR coverage.

The results support a measurable but limited benefit from classical LiDAR–RADAR fusion and identify concrete work for Phase 3.